

Project Name: FarmFreshDirect	Tech Stack: React 19, Vite, Express, Node, MongoDB, JWT
Architecture: RESTful SPA + Best-Seller Sorting Engine	Security: Bcrypt, JWT Auth, Resend 6-Digit OTP
Author/Team: Engineering Team	Date: July 2026

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2. Comprehensive Technology Stack Breakdown

Tier	Technologies & Libraries	Primary Responsibility
Frontend Core	React 19, React-DOM, Vite 8	Component rendering, state management, near-instant HMR dev bundling.
Routing & UI	React Router DOM v6, Lucide React	Client-side SPA navigation (/farmer, /consumer) & scalable vector icons.
Best Seller Engine	soldCount Index & Sorting	Prioritizes high-demand produce so highest selling items render at top.
Animations	GSAP 3.15, HTML5 Canvas API	Staggered timeline reveals, 3D perspective mouse tilt, particle physics canvas.
Backend API	Node.js, Express.js 4.19, CORS	REST API routing, middleware execution, HTTP request lifecycle control.
Database	MongoDB, Mongoose 8.4, Custom JSONModel	NoSQL BSON document persistence with automatic offline JSON fallback adapter.
Authentication	JSONWebToken (JWT) 9.0, BcryptJS 3.0	Stateless signed authorization headers & 10-round password salt hashing.
Email Services	Resend API 6.12	Transactional HTML OTP delivery for signups and password recovery.

3. Best-Seller Ranking Engine & Page Walkthrough

A. How Best-Seller Priority Ranking Operates:

- Data Tracking (soldCount):** Every product schema tracks a soldCount attribute representing total cumulative units ordered.
- Atomic Sales Increments:** When a customer checks out, the backend executes an atomic database operation: \$inc: { quantity: -N, soldCount: N }.
- Default Primary Sort:** Backend queries and frontend state sort produce by soldCount: -1 (descending), ensuring top-selling items render FIRST in the grid.
- Best Seller Badges:** Highlights top sellers with a ■ **BEST SELLER** gradient badge and displays individual sales counts (e.g. ■ 45 kg sold).

B. Consumer Marketplace Page Walkthrough:

- **Default Priority Grid:** Products selling the most appear at the very top of the catalog.
- **Sort Selector:** Includes a sort control allowing switching between ■ Best Sellers (Most Sold), ■ Newest Harvest, ■ Price: Low to High, and ■ Price: High to Low.
- **Details Modal:** Shows total units sold alongside available stock so buyers see real-time community demand.

Documentation updated automatically for FarmFreshDirect application workspace.